

- 3 The photograph shows a species of mayfly nymph. Mayfly nymphs live in streams and rivers on every continent except Antarctica.



(Source: © NaturePics / Alamy Stock Photo)

Many species of mayfly nymph are found in unpolluted streams.

All mayfly species have a three-pronged tail.

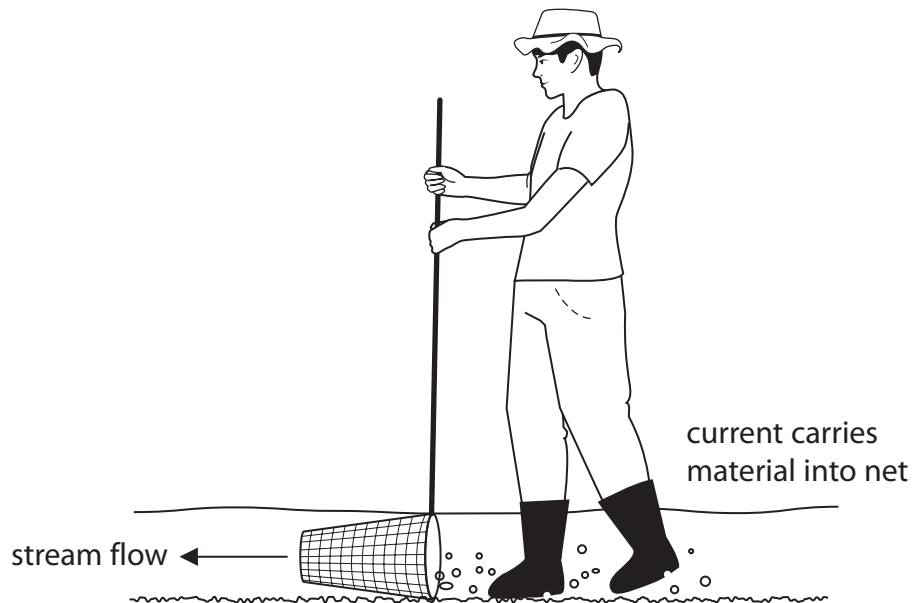
There are fewer species and lower numbers of each species in polluted streams.

A student read a report that low levels of chlorpyrifos (an insecticide) may have entered some streams.

The student used the method shown in the diagram to collect mayfly nymphs.

Samples were collected from unpolluted stream **A** and polluted stream **B**.

The diagram shows kick sampling in a stream.



The number of mayfly nymphs from each kick sample was recorded.

The student's results:

Stream **A**

27 37 24 45 34 38 49 61 40 20 28 38 42 25 32

Stream **B**

25 16 34 12 35 26 43 51 26 24 36 26 49 52 19

(a) State a suitable null hypothesis for this investigation.

(1)

.....

.....

.....

.....

(b) Draw a suitable table to display the **raw data** and your calculated **means** for stream **A** and stream **B**.

(3)

- (c) Plot a graph to show the mean number of mayfly nymphs for stream **A** and stream **B**.

Include an indication of the variability of the data.

(3)



(d) The student analysed the data using a t test with the formula:

$$t = \frac{(\bar{x}_A - \bar{x}_B)}{\sqrt{\frac{(S_A)^2}{n_A} + \frac{(S_B)^2}{n_B}}}$$

where:

$\bar{x}$ is the mean value for each stream

n is the number of samples for each stream

$(S_A)^2 = 116$ and $(S_B)^2 = 160$

(i) Calculate the value of t .

(3)

Answer

(ii) The table shows the critical values of t for different degrees of freedom.

The number of degrees of freedom = $(n_A - 1) + (n_B - 1)$

Degrees of freedom	p = 0.05	p = 0.01
15	2.13	2.95
16	2.12	2.92
17	2.11	2.90
18	2.10	2.88
19	2.09	2.86
20	2.09	2.84
21	2.08	2.83
22	2.07	2.82
23	2.07	2.81
24	2.06	2.80
25	2.06	2.79
26	2.06	2.78
27	2.05	2.77
28	2.05	2.76
29	2.04	2.76
30	2.04	2.75

Comment on the results of this investigation.

Use your graph and the information in the table to support your answer.

(3)

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins or other markings on the paper.

(e) Explain why it may **not** be reasonable to draw valid conclusions from the results of this investigation.

(2)

(Total for Question 3 = 15 marks)
